PARENT ROUNDTABLE SERIES

The Science Behind Inflammatory Foods

How gut inflammation becomes whole-body inflammation, and the everyday food choices that shift it

Deborah M. Burke-Allen, RPh

Registered Pharmacist | Parent education roundtable



THE BIG PICTURE

Inflammation has many sources; food is the one families shape every day



- Symptoms often reflect the total load, not one single cause
- Genetics are fixed; food, environment, and stress can change
- When the load drops, symptoms often settle

CHILDREN ARE NOT JUST LITTLE ADULTS

Early signs of inflammation show up across the whole body

Skin

- Diaper rash
- Rash around the mouth
- Cradle cap
- Sensitive skin

Behavior

- Fussiness
- Colic
- Disturbed sleep

Gut

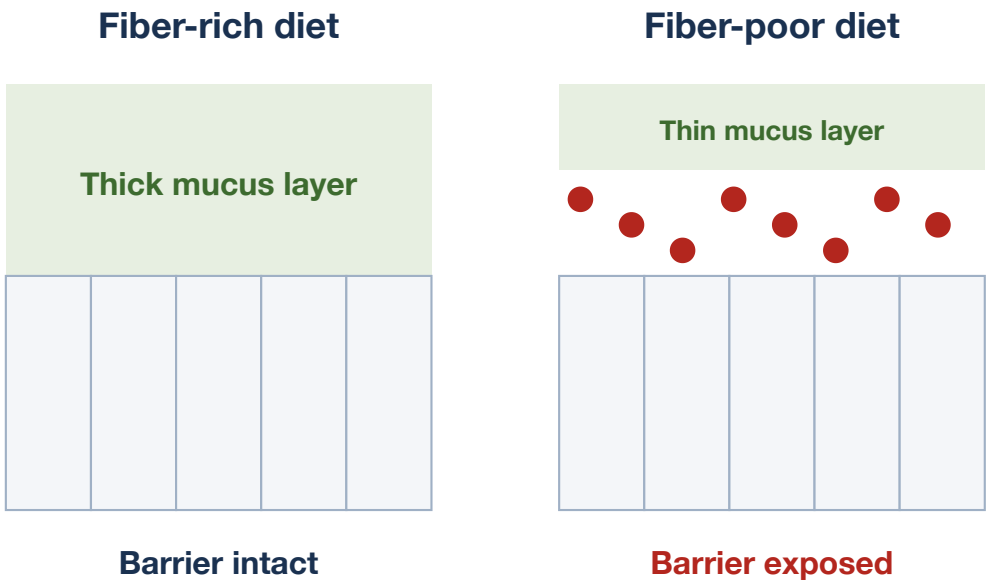
- Bloated tummy
- Diarrhea
- Constipation

Frequent illness

- Congestion
- Ear infections

Each sign has many possible causes. Together, they are a reason to look closer, not a diagnosis.

Feed the gut, or it will feed on you

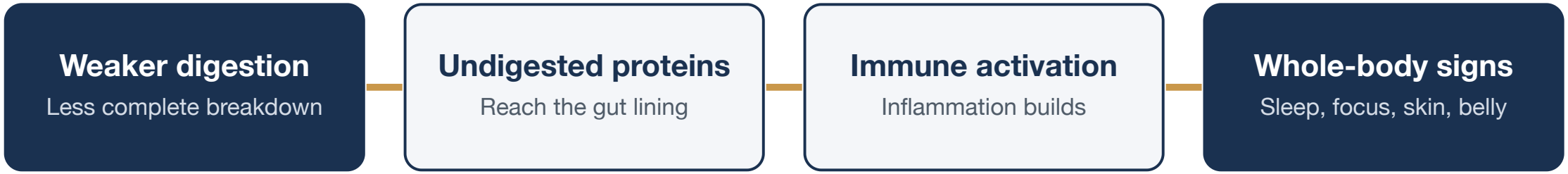


- Fiber feeds the gut microbes that help keep the protective mucus layer thick
- In a mouse study, microbes starved of fiber began breaking down that mucus layer instead
- Vegetables, fruits, beans, and whole grains keep the barrier fed

Desai MS, et al. Cell. 2016;167(5):1339-53. PMID 27863247 (animal study)

THE RIPPLE EFFECT

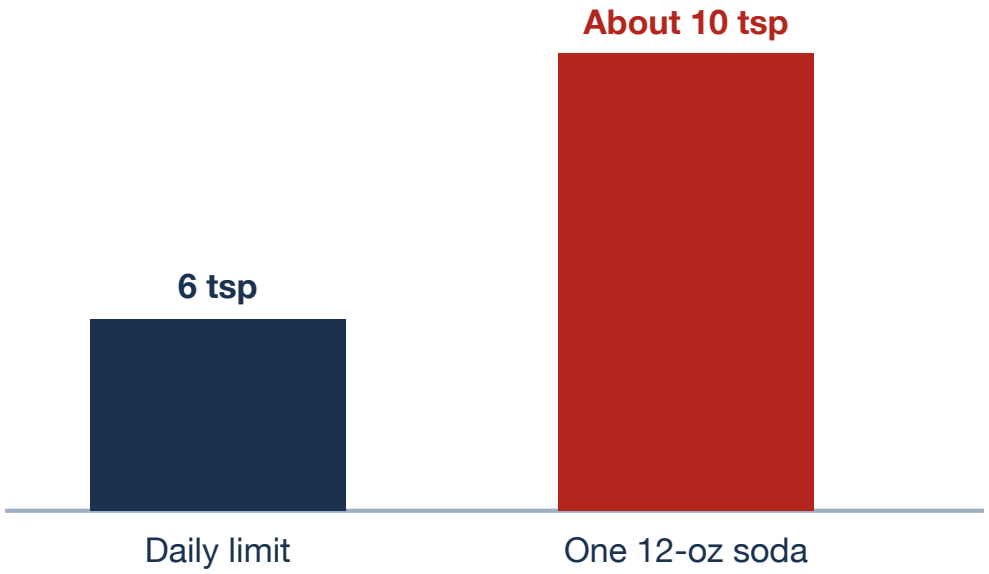
When digestion struggles, the effects ripple outward



This is why we start with the gut: calming digestion often eases symptoms that seem unrelated, from congestion to restless sleep.

ADDED SUGAR

Children should have less than 6 teaspoons of added sugar a day



- The American Heart Association advises under 25 grams (about 6 teaspoons) of added sugar a day for ages 2 to 18
- No added sugar before age 2
- No more than one 8-ounce sugary drink a week

Vos MB, et al. Circulation. 2017;135(19):e1017-34. PMID 27550974

WHY THE MESSAGE WAS SLOW

Industry funding shaped the sugar story for decades

- Internal documents show a sugar industry group funded 1960s research that played down sugar's role in heart disease
- The research shifted attention to dietary fat instead
- That framing influenced dietary advice for years

The lesson for families

Ask who funded the study. Independent research and current guidelines are a better guide than marketing.

Kearns CE, et al. JAMA Intern Med. 2016;176(11):1680-5. PMID 27617709

Artificial food dyes can affect attention and activity in some children

- In a randomized, controlled trial, drinks with common artificial colors and a preservative increased hyperactivity in children from the general population
- Since 2010, foods sold in the European Union that contain six of these dyes must carry a warning about effects on activity and attention

Practical swaps

- Read labels for numbered colors (for example, Red 40, Yellow 5, Yellow 6)
- Choose snacks colored with foods such as beet or turmeric
- One treat often contains several dyes at once

McCann D, et al. Lancet. 2007;370(9598):1560-7. PMID 17825405

A supervised elimination diet can help some children with ADHD

- In a randomized trial, children on a strict, supervised elimination diet had a large drop in ADHD symptoms compared with controls
- Symptoms returned in many children when trigger foods were added back
- Blood IgG food tests did not predict which foods caused the relapse

Do it safely

- Plan it with a clinician or dietitian
- Keep the diet nutritionally complete
- Add foods back one at a time and track the response

Pelsser LM, et al. Lancet. 2011;377(9764):494-503. PMID 21296237

DAIRY

Dairy: lactose, casein, and the A1 and A2 question

Lactose

The sugar in milk. Lactose intolerance causes gas, bloating, and loose stools.

Casein

The main milk protein. Some children react to the protein itself, regardless of lactose.

A1 and A2

Digesting A1 beta-casein can release a peptide called BCM-7; A2 does not. Early evidence links A1 to digestive discomfort in some people.

> If a child improves off cow's milk, casein sensitivity is possible with both A1 and A2 milk. Remove dairy gradually, with a plan for calcium and vitamin D.

Brooke-Taylor S, et al. Adv Nutr. 2017;8(5):739-48. PMID 28916574

ORGANIC CHOICES

Choosing organic produce lowers children's pesticide exposure

- When children switched to organic foods, pesticide byproducts in their urine dropped to undetectable or near-undetectable levels within days
- Children with higher levels of these byproducts had higher odds of ADHD, an association that does not prove cause

Spend where it counts

Prioritize organic for the produce most likely to carry residues, such as strawberries and spinach, and buy conventional for thick-skinned produce such as avocados.

THE HEALTHY PLATE

A better plate lowered heart disease risk by up to 39%

- Adults who ate closest to the Harvard-style healthy eating pattern had 39% lower cardiovascular disease risk in men and 28% lower in women

> Major chronic disease risk was 20% lower in men and 11% lower in women.

What the plate looks like

- Water instead of sugary drinks
- More vegetables than fruit; potatoes do not count
- Whole grains over refined grains
- Fish, poultry, beans, and nuts
- Olive oil; avoid trans fats

McCullough ML, et al. Am J Clin Nutr. 2002;76(6):1261-71. PMID 12450892

WHERE TO START

Start small: one week of simple swaps

Remove or reduce	Sugary drinks	Artificial colors	Packaged snacks	Refined sugar	Processed meats
Replace with	Filtered water	Green vegetables	Real-food snacks	Naturally sweet foods	Clean proteins

One change at a time

Small, steady swaps stick better than an overnight overhaul.

Expect a healing curve

Like a sprained ankle, an irritated gut can take months to fully settle.

Test later, not first

Food sensitivity testing is clearer after the gut has calmed and the noise is lower.

REFERENCES

Sources

1. Desai MS, et al. Cell. 2016;167(5):1339-53. PMID 27863247. Dietary fiber-deprived gut microbiota degrades the colonic mucus barrier.
2. Vos MB, et al. Circulation. 2017;135(19):e1017-34. PMID 27550974. Added sugars and cardiovascular disease risk in children.
3. Kearns CE, et al. JAMA Intern Med. 2016;176(11):1680-5. PMID 27617709. Sugar industry and coronary heart disease research.
4. McCann D, et al. Lancet. 2007;370(9598):1560-7. PMID 17825405. Food additives and hyperactive behaviour in children.
5. Pelsser LM, et al. Lancet. 2011;377(9764):494-503. PMID 21296237. Restricted elimination diet in children with ADHD (INCA study).
6. Brooke-Taylor S, et al. Adv Nutr. 2017;8(5):739-48. PMID 28916574. Gastrointestinal effects of A1 compared with A2 beta-casein.
7. Lu C, et al. Environ Health Perspect. 2006;114(2):260-3. PMID 16451864. Organic diets lower children's exposure to organophosphorus pesticides.
8. Bouchard MF, et al. Pediatrics. 2010;125(6):e1270-7. PMID 20478945. ADHD and urinary metabolites of organophosphate pesticides.
9. McCullough ML, et al. Am J Clin Nutr. 2002;76(6):1261-71. PMID 12450892. Diet quality and major chronic disease risk.